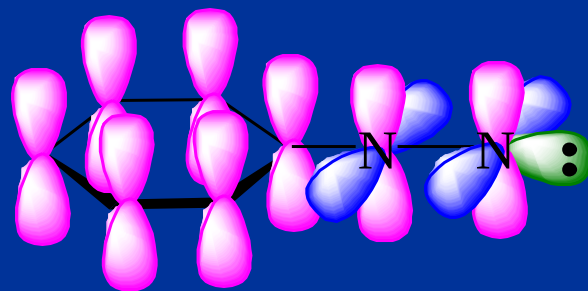
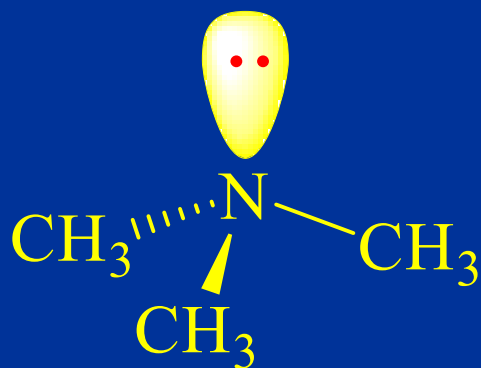


第十四章 含氮化合物

Nitrogen Compounds

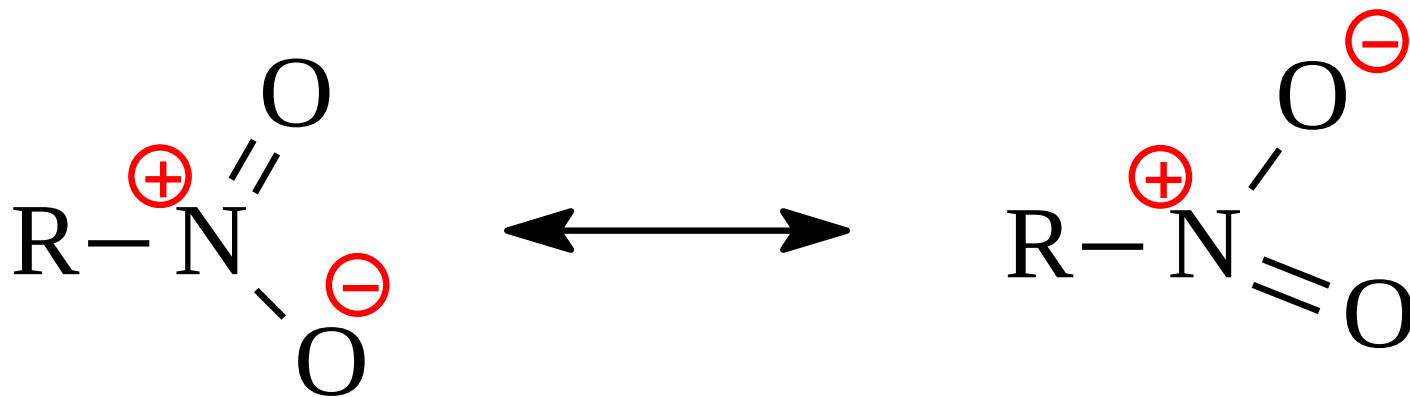


CONTENT

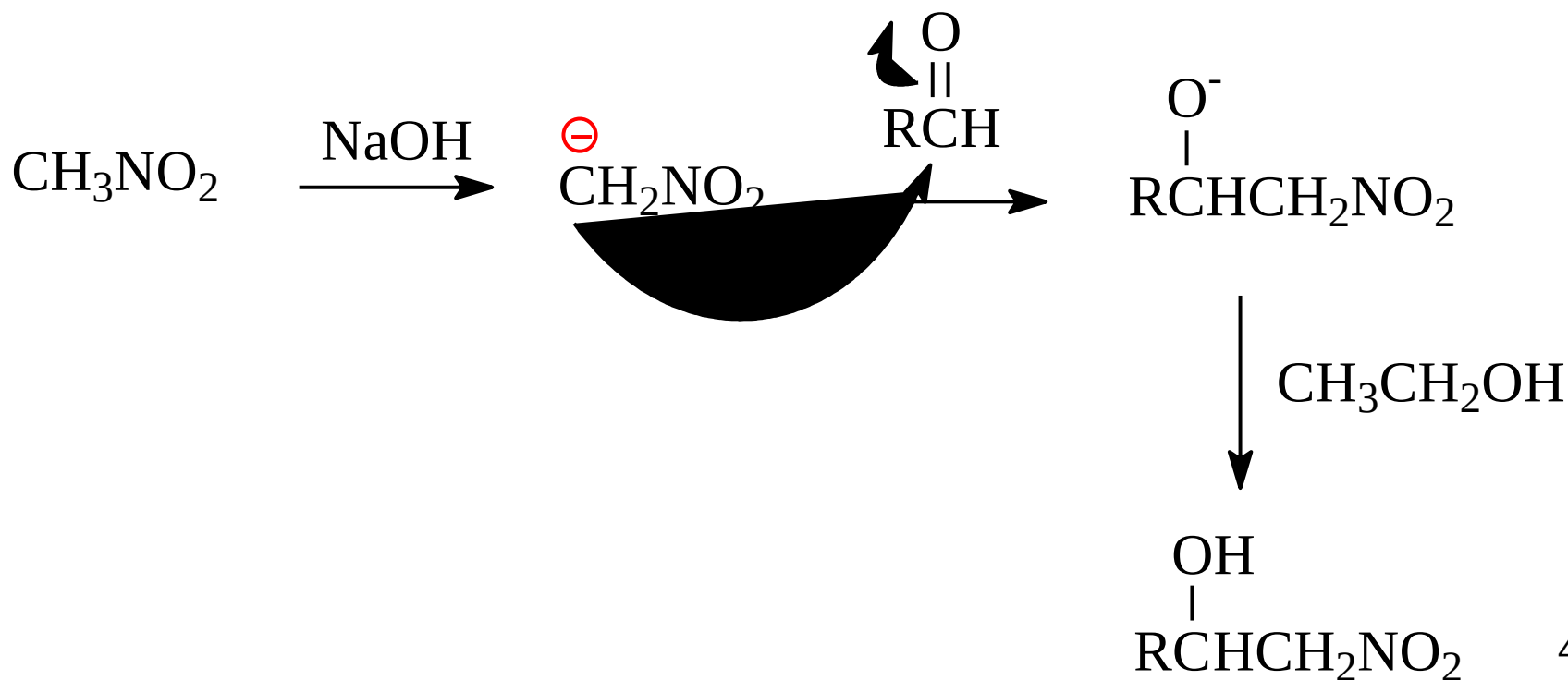
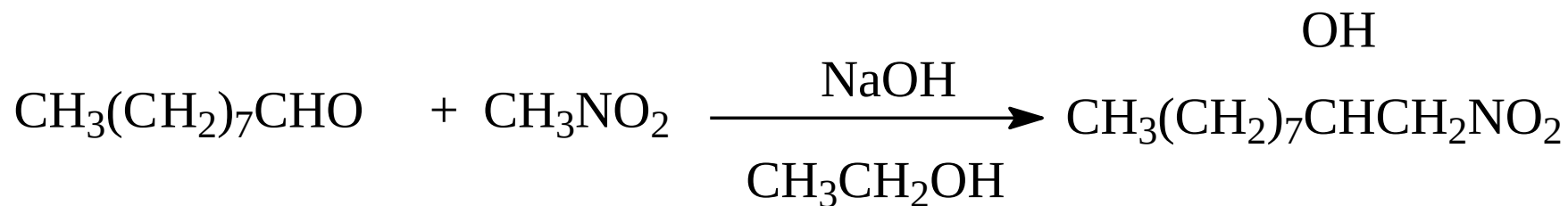
- 1 硝基化合物
- 2 胺的结构和物理性质
- 3 胺的制备
- 4 胺的化学性质
- 5 季铵盐化合物
- 6 重氮与偶氮化合物

硝基化合物

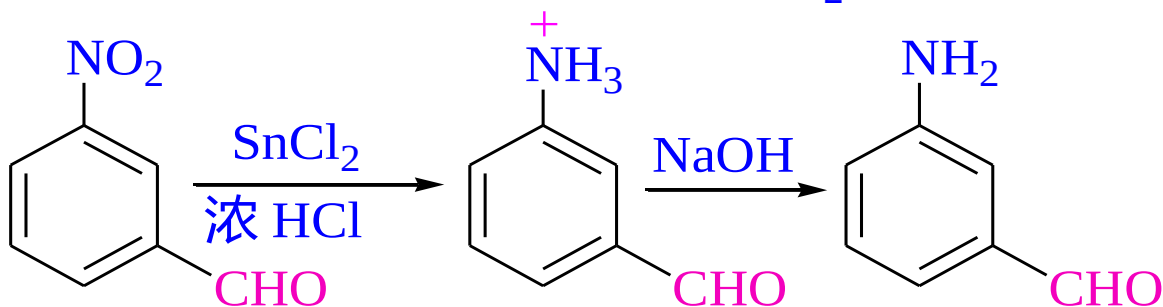
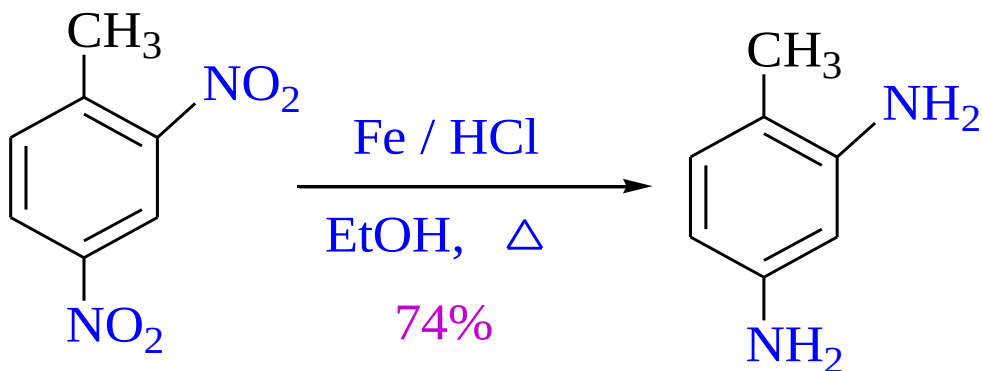
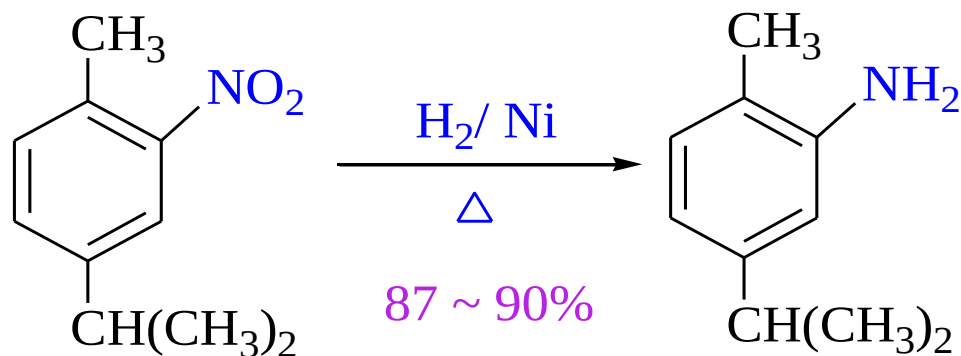
硝基化合物的结构(了解)



α -H 的反应

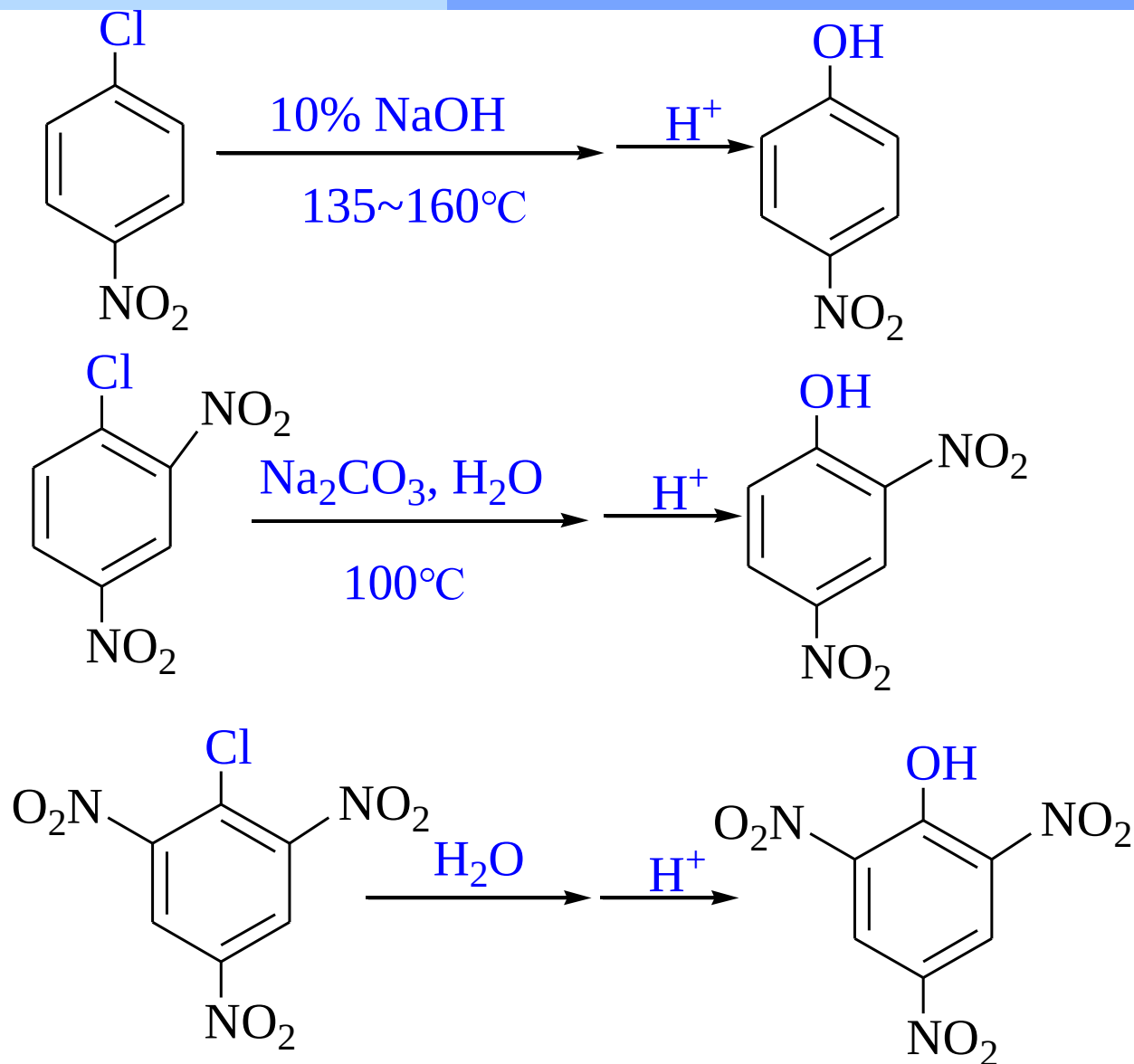


硝基化合物的还原反应



常用
 Fe / HCl
 Zn / HCl

硝基对氯苯水解影响

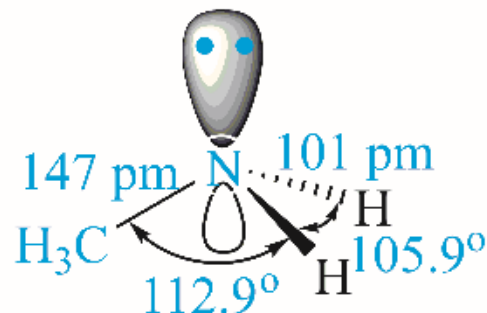
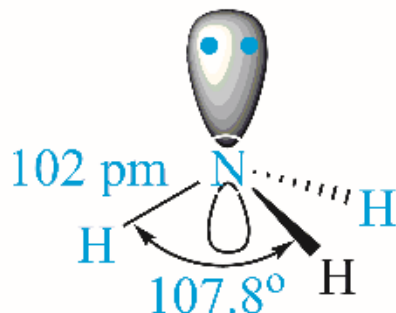


亲核取代

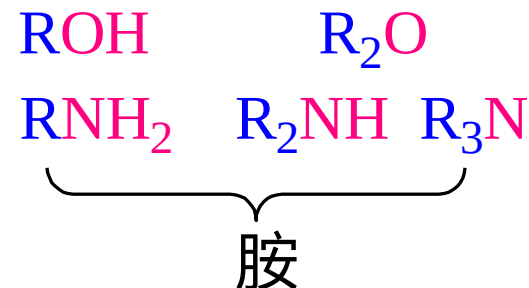
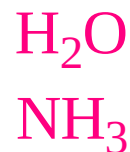
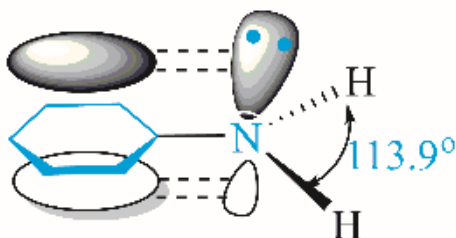
硝基对苯环上邻对位取代基的影响

胺的结构和物理性质

• Structures of aliphatic amines



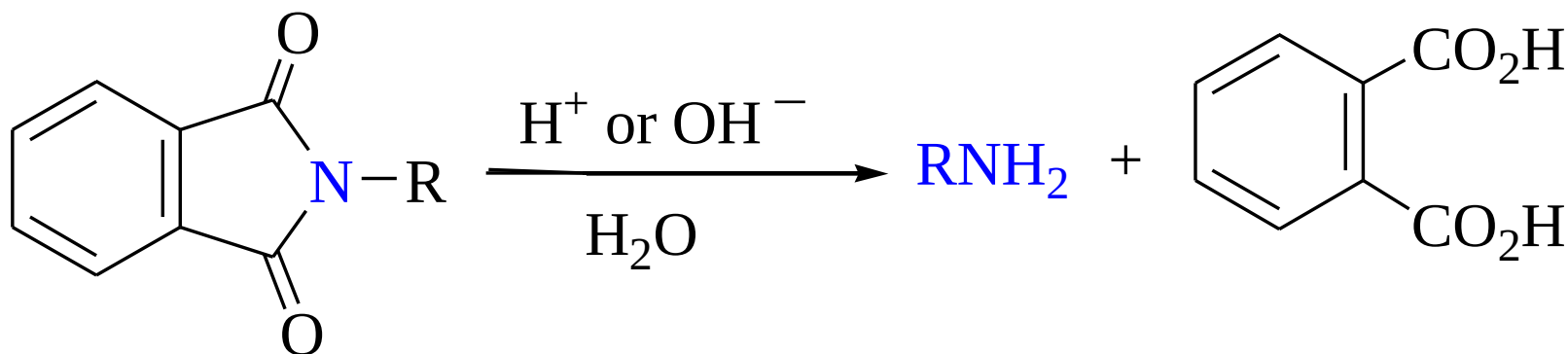
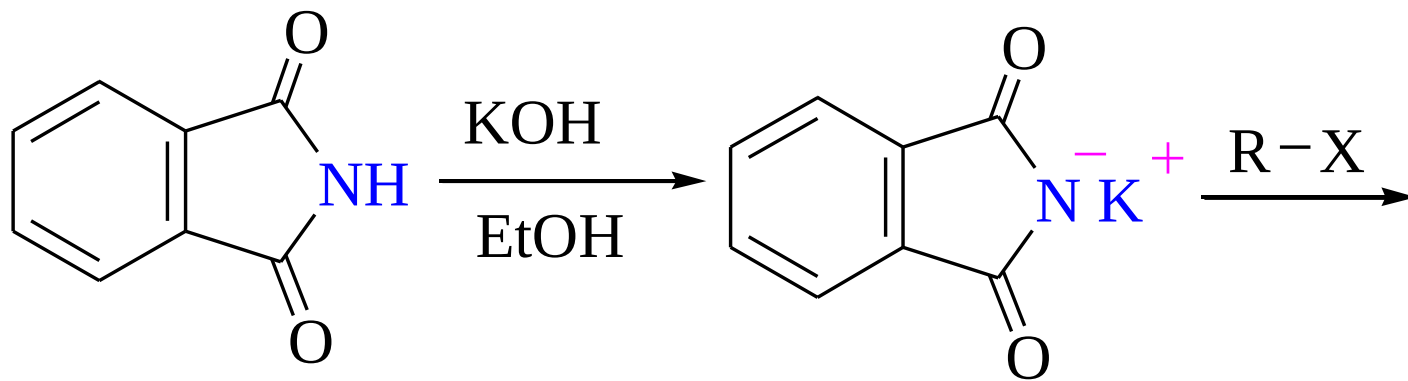
• Structures of aromatic amines



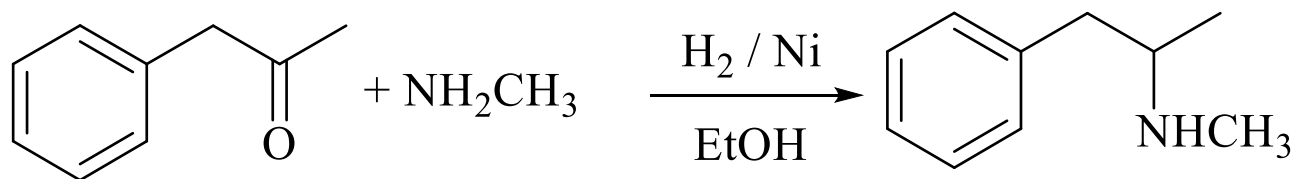
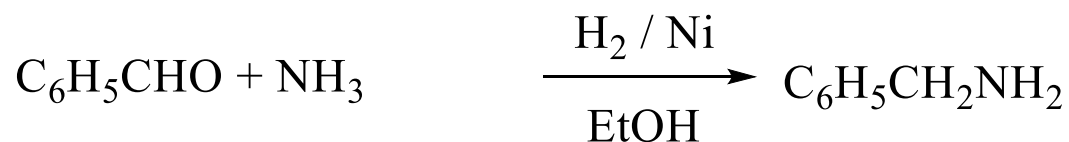
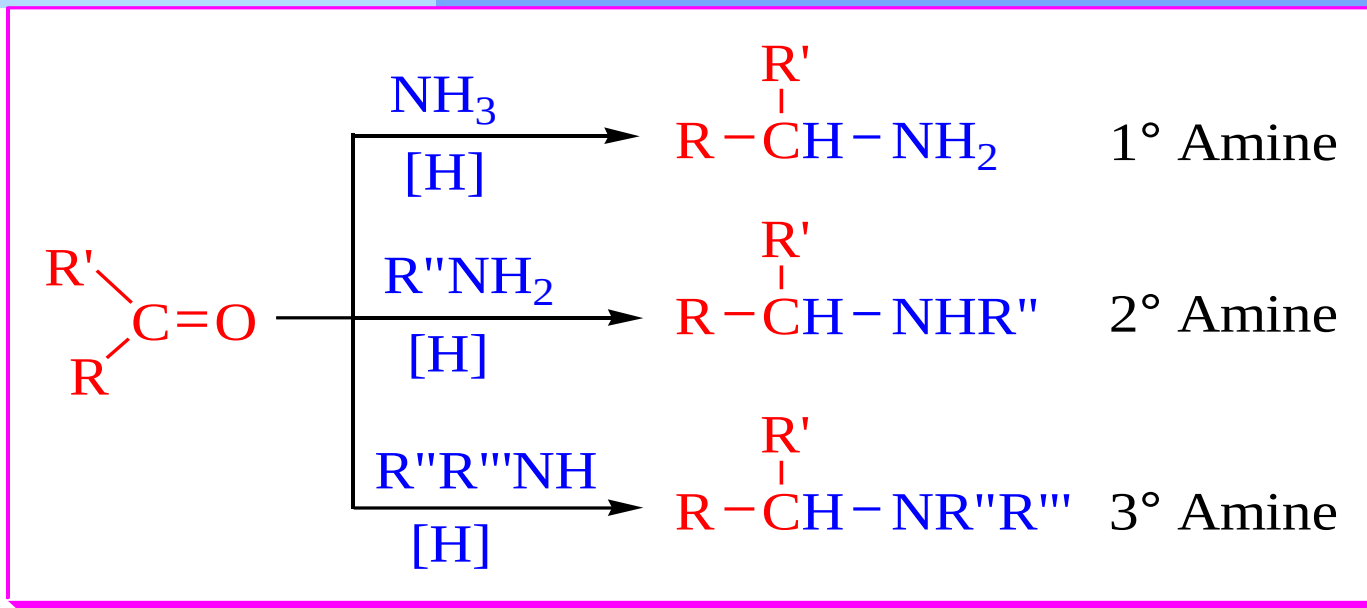
伯胺 仲胺 叔胺
(一级胺) (二级胺) (三级胺)

胺的制备

Gabriel 合成法



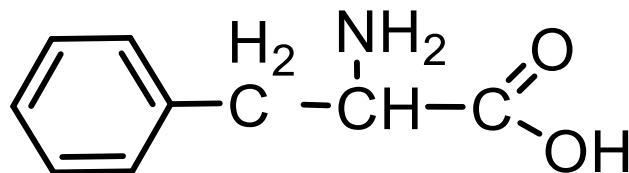
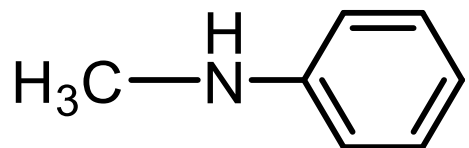
醛和酮的氨化还原



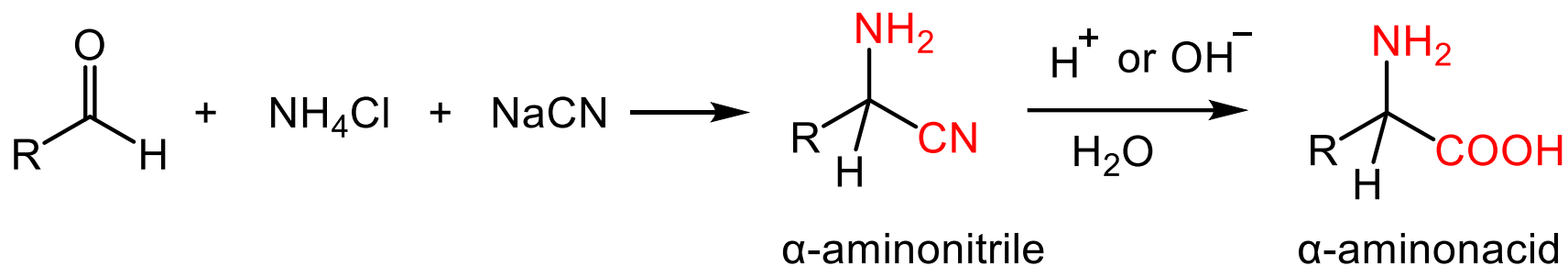
其它制备方式请自学

测试

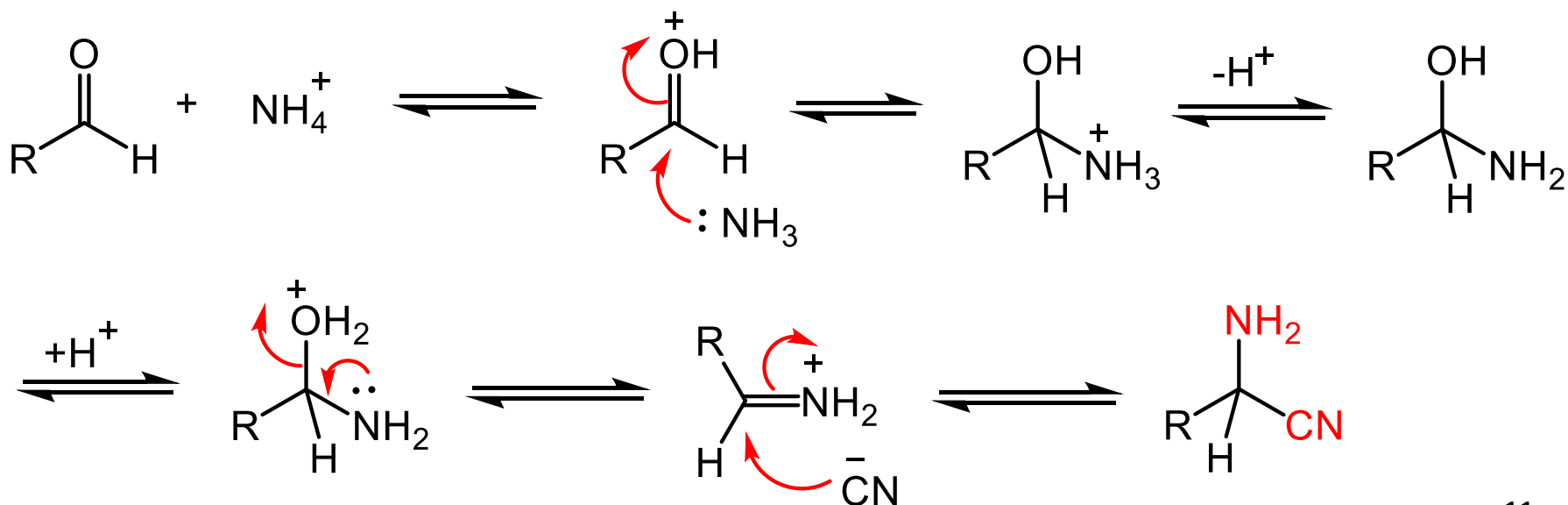
1) 以苯、甲苯以及不超过3个碳的有机试剂合成下列物质



羰基化合物与氯化铵、氰化钠反应，形成 α -氨基腈，经水解可以制备 α -氨基酸：

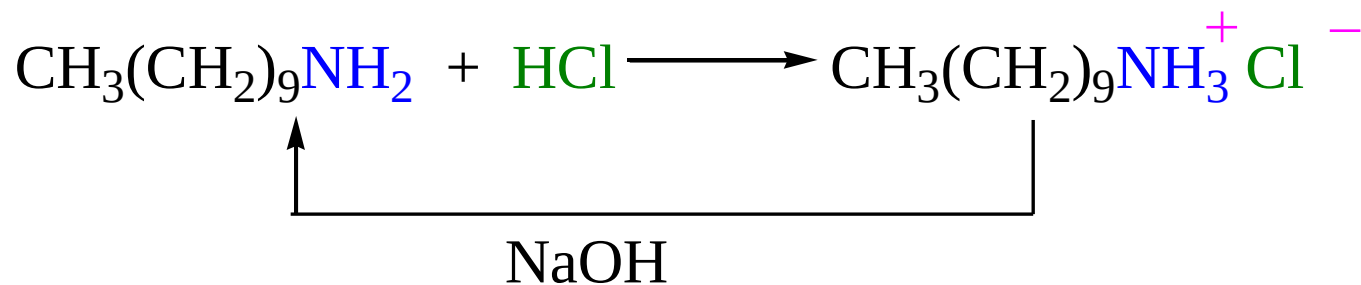
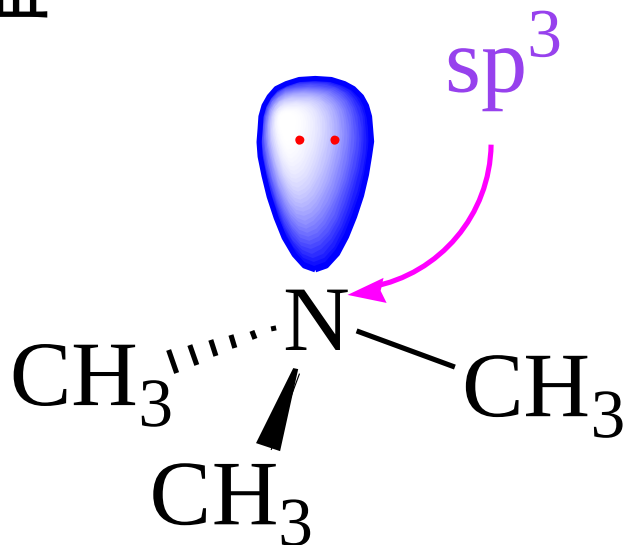


Mechanism:



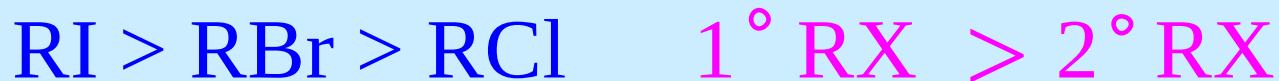
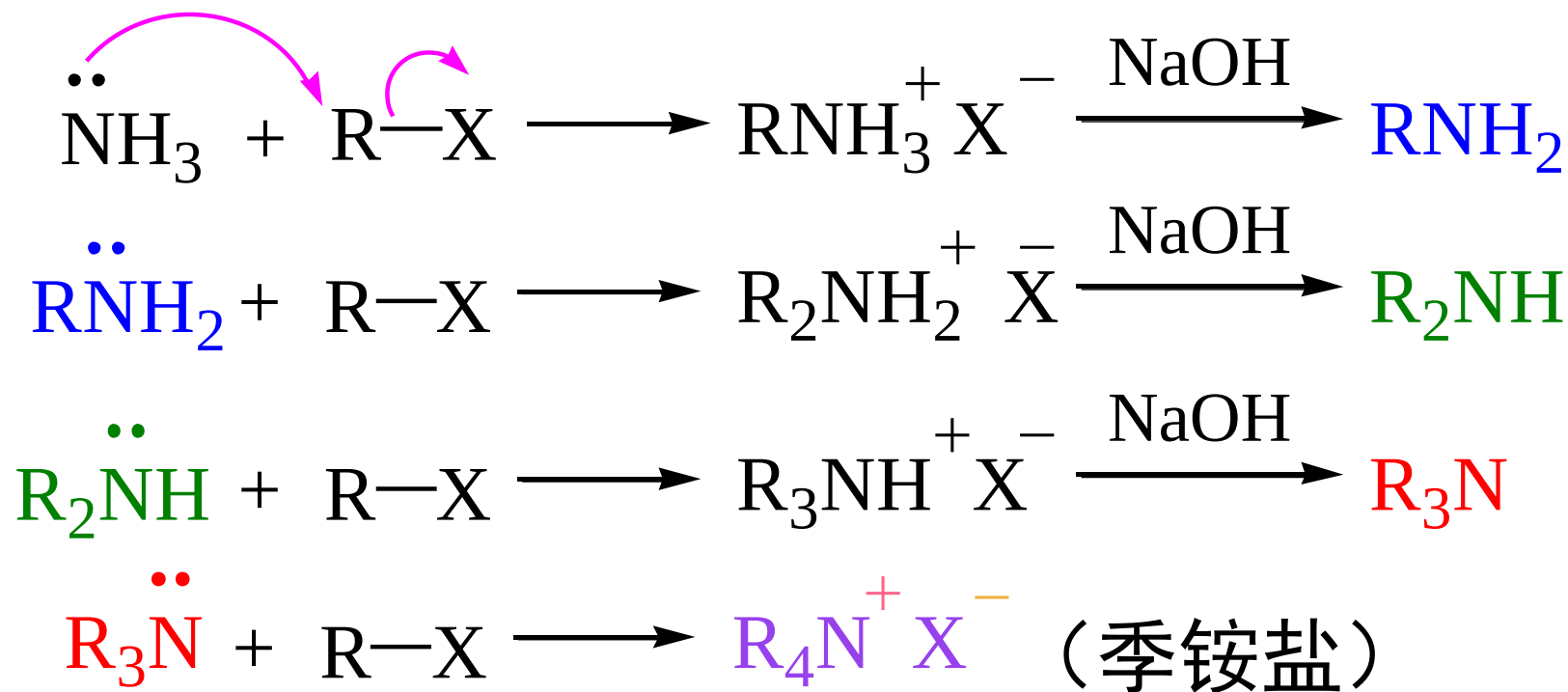
胺的化学性质

胺的碱性和成盐



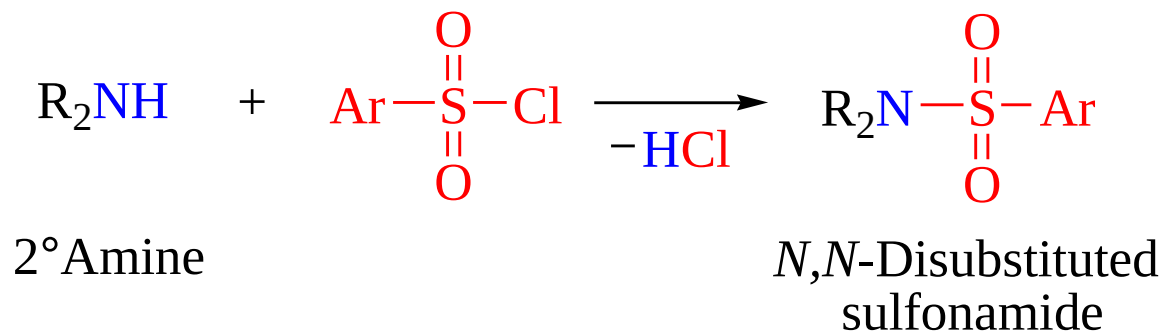
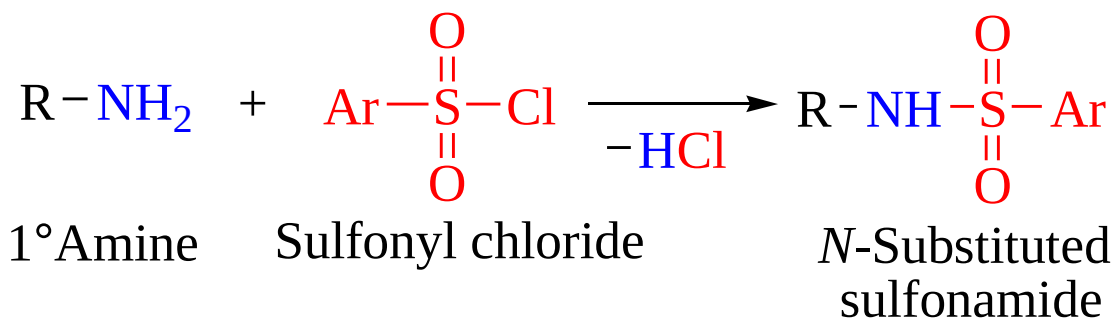
胺的烃基化

为什么不能用氯代烃直接制备伯仲叔胺？



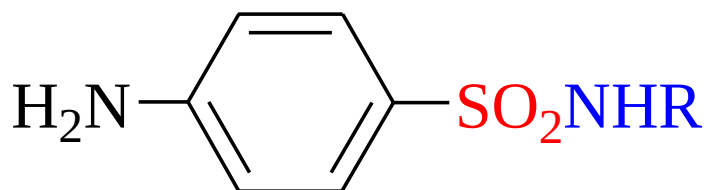
磺酰胺

Primary and secondary amines react with sulfonyl chlorides to form **sulfonamides**.



磺酰胺药物

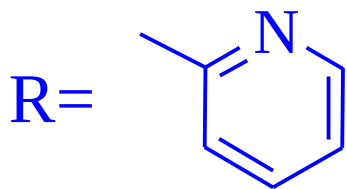
Sulfa Drugs



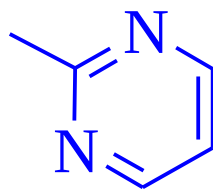
Sulfonilamide

(a sulfa drug)

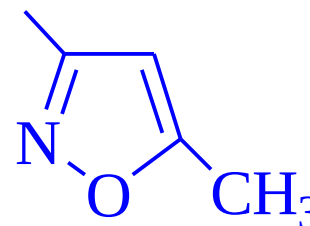
对氨基苯磺酰胺



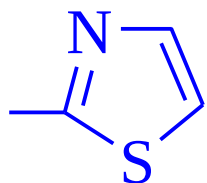
Sulfapyridine
磺胺吡啶(SP)



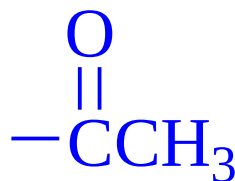
Sulfadiazine
磺胺嘧啶 (SD)



Sulfamethoxazole
磺胺甲基异噁唑(SMZ)



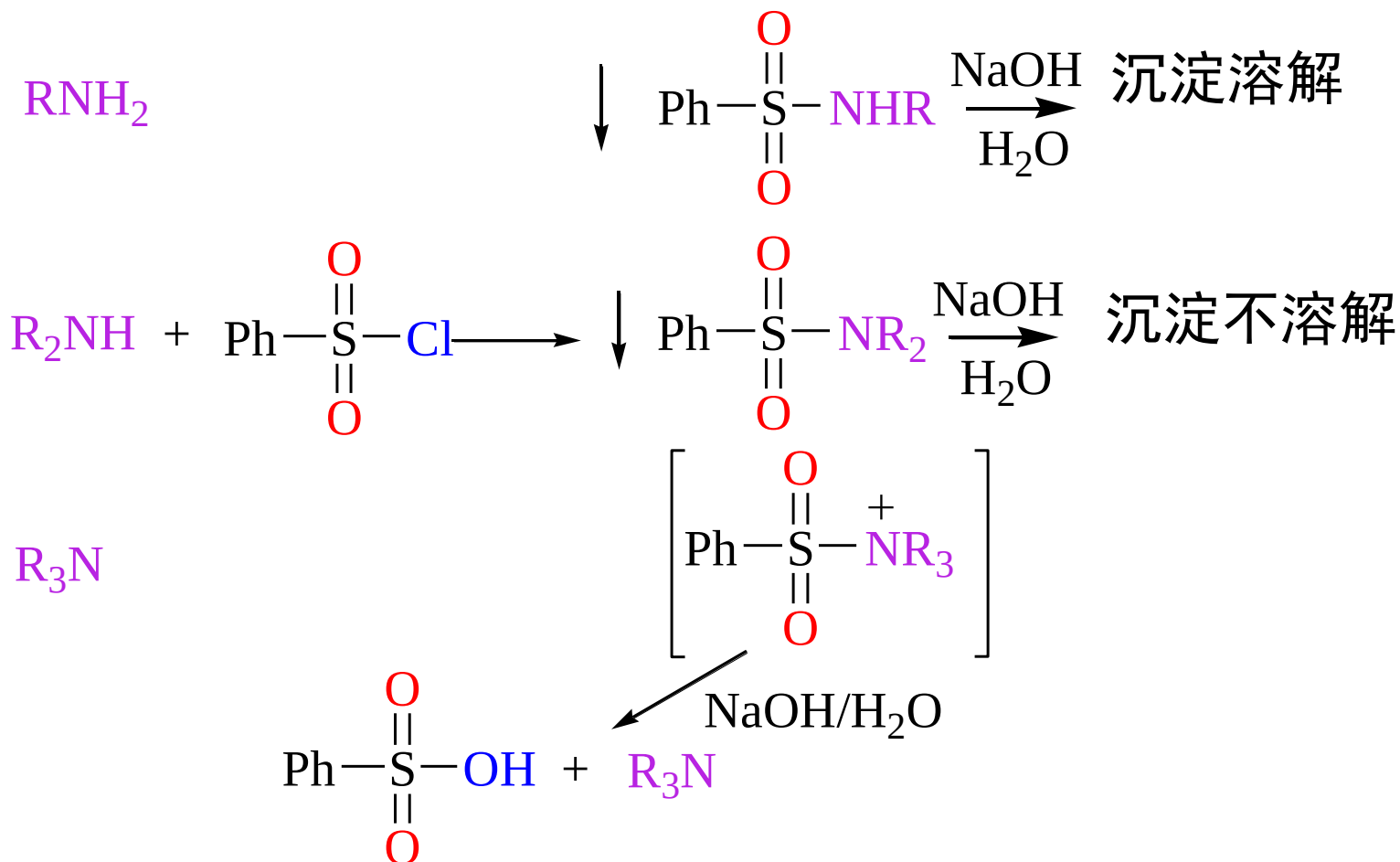
Sulfathiazole
磺胺噻唑(ST)



Sulfaacetamide
对乙酰氨基苯磺酰胺 (SA)

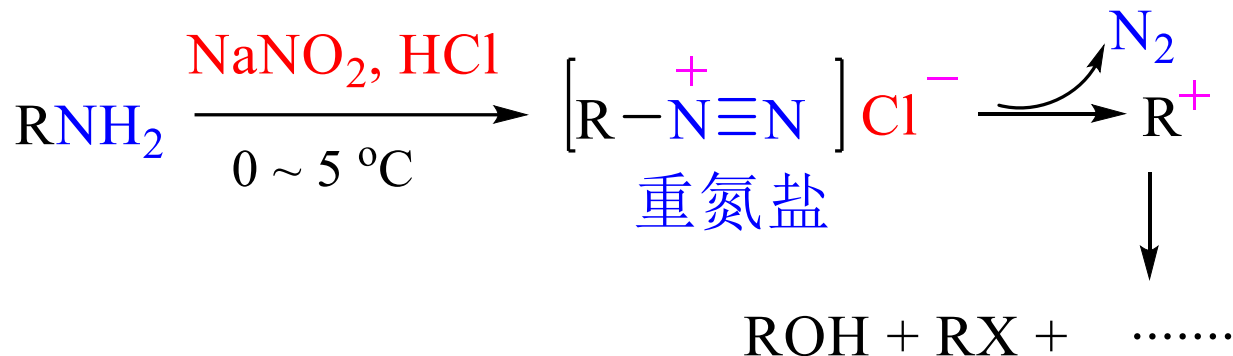
胺的分离鉴定

胺的磺酰化： 用于鉴别分离伯胺仲胺和叔胺

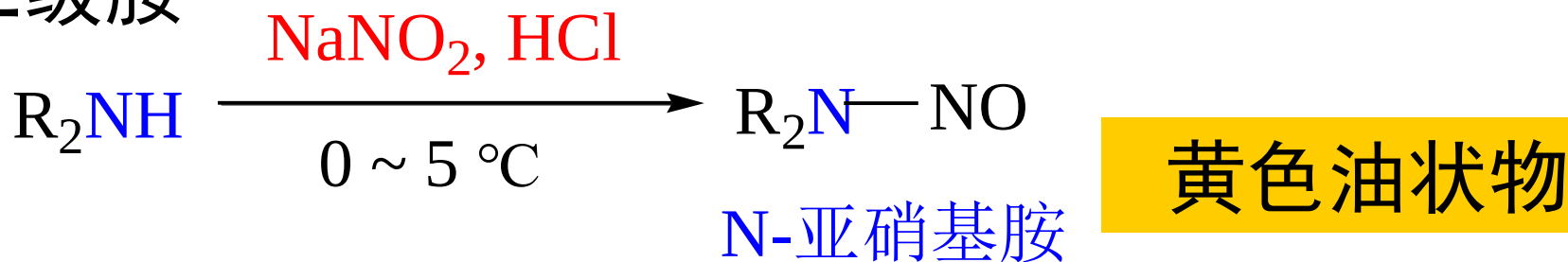


脂肪族胺与亚硝酸反应

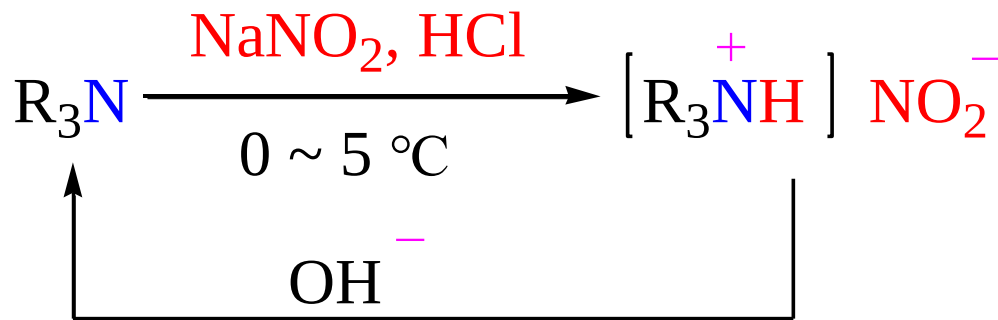
一级胺



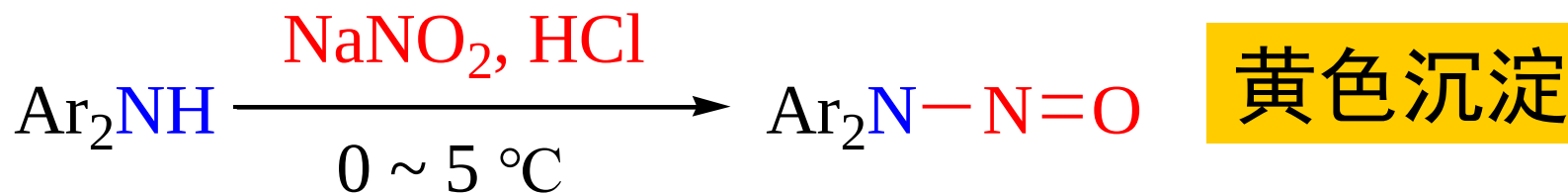
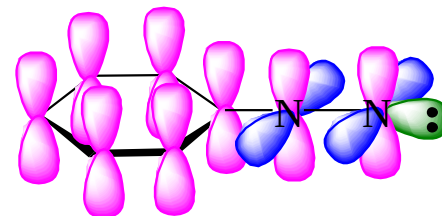
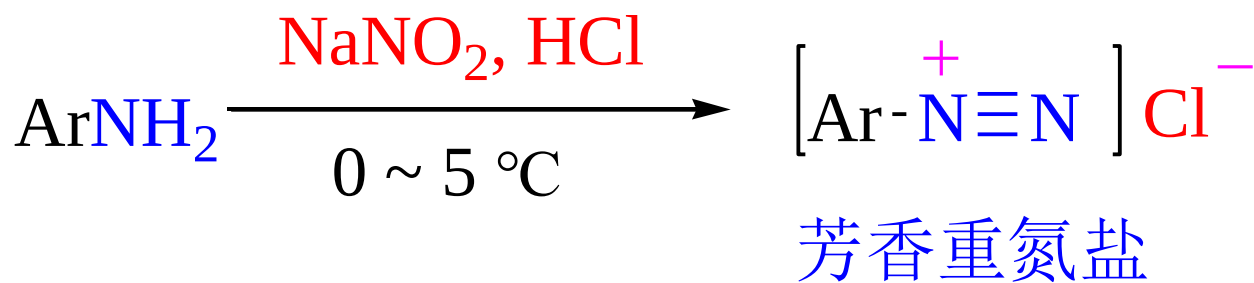
二级胺



叔胺



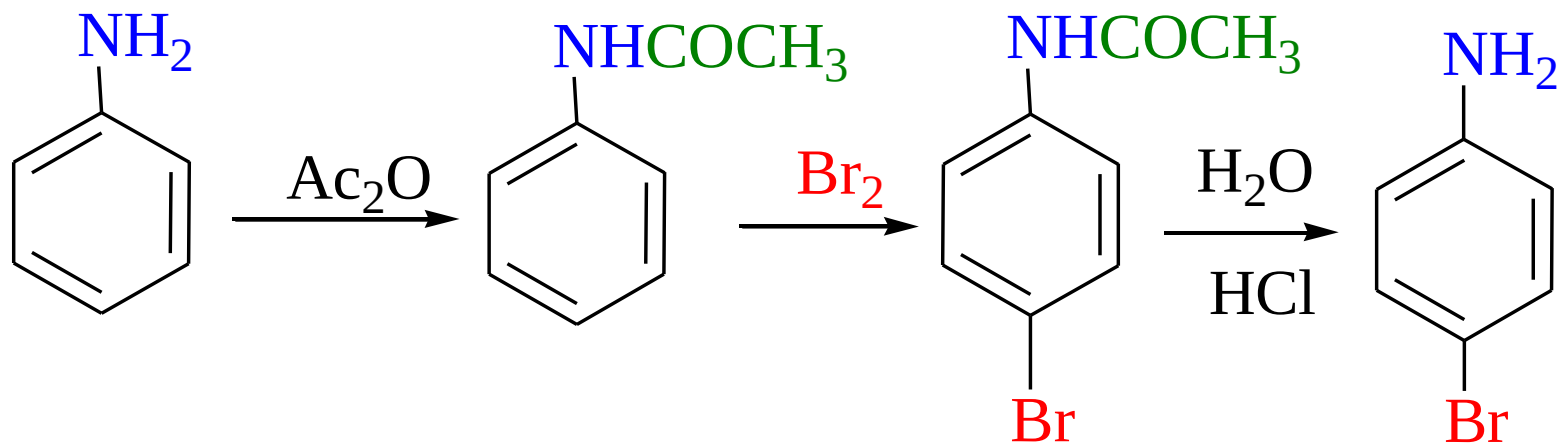
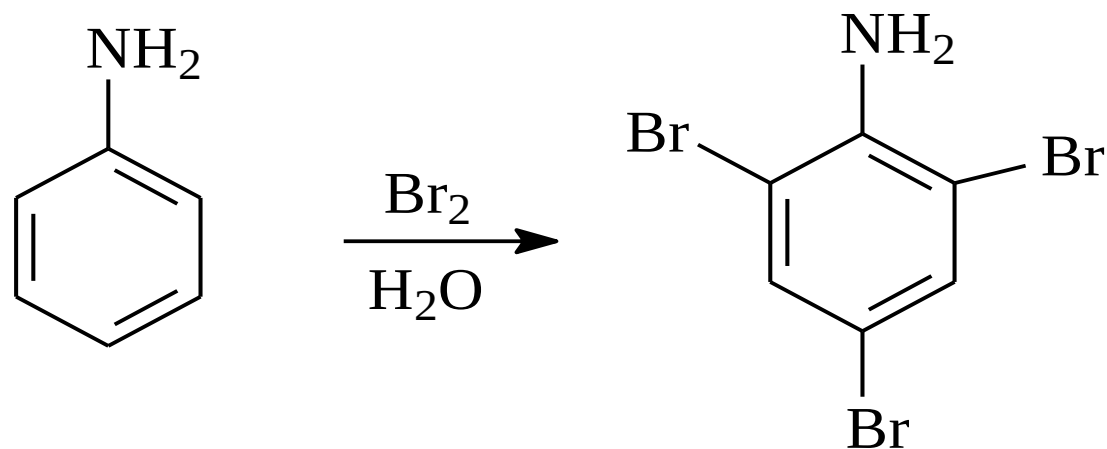
芳香胺与亚硝酸反应



绿色晶体

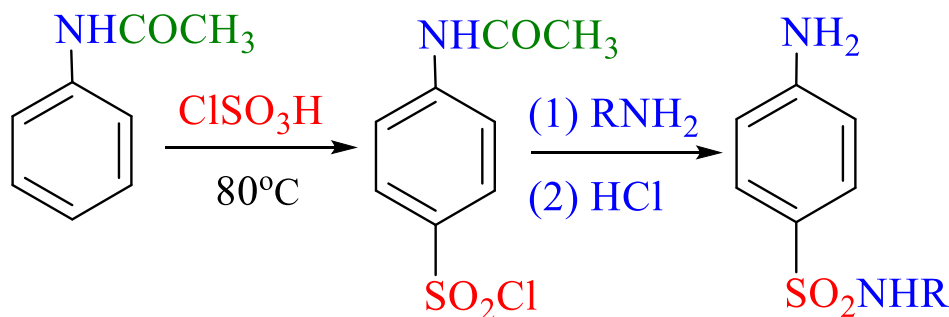
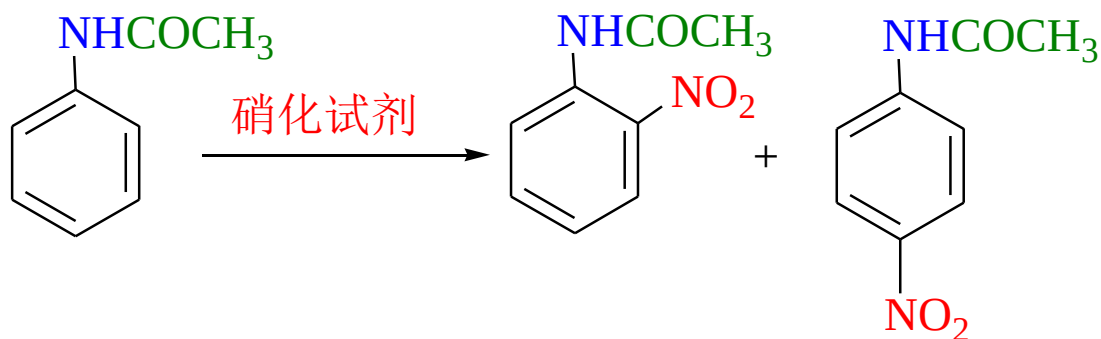
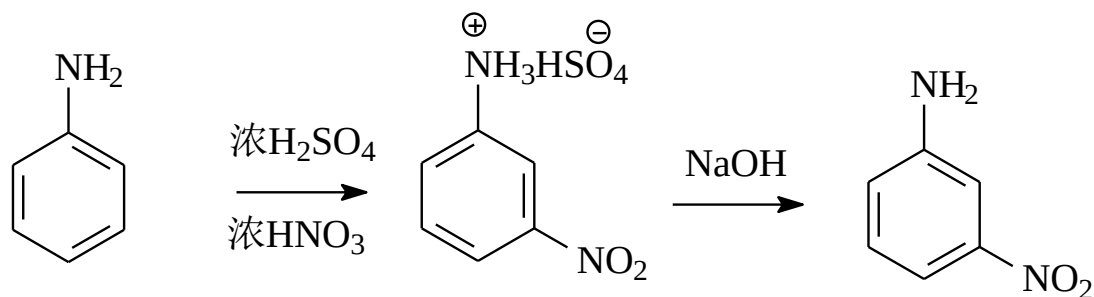
芳胺的亲电取代反应

卤化反应

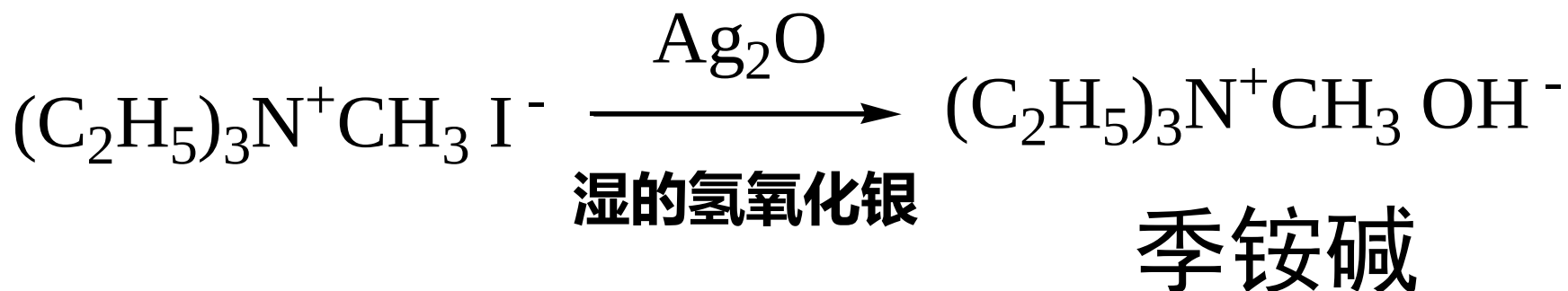
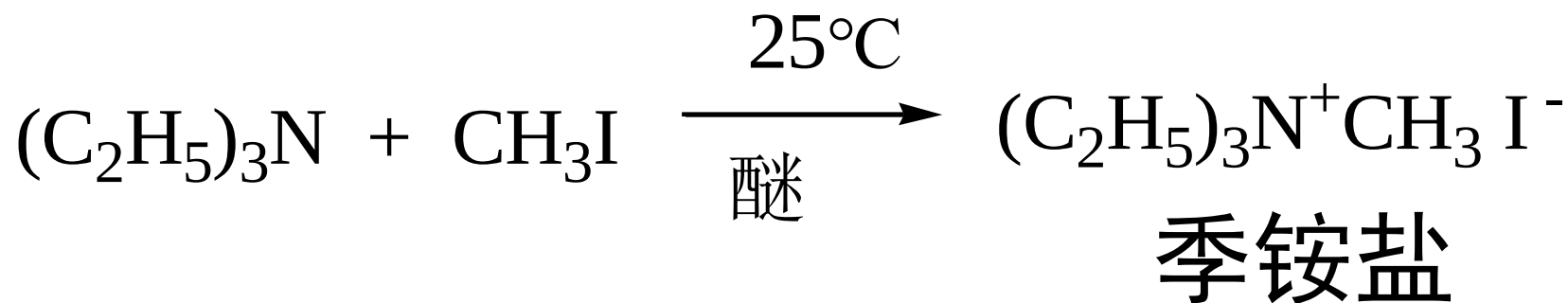


芳胺硝化、磺化反应

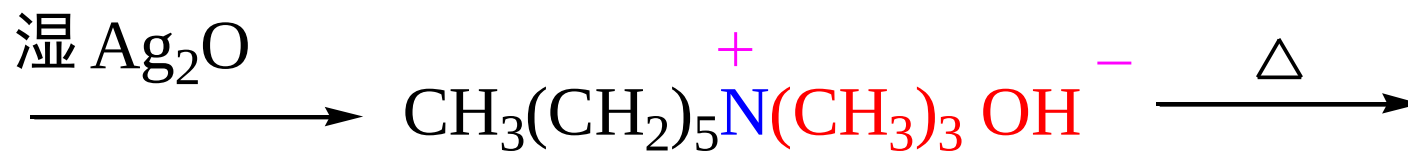
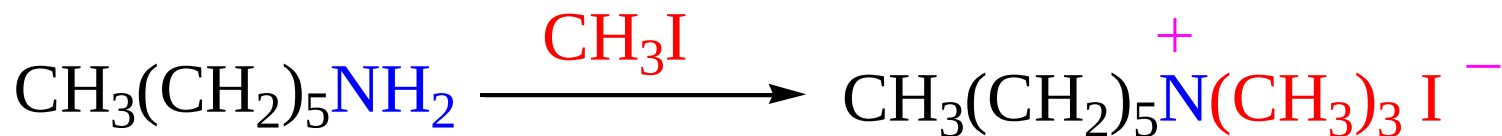
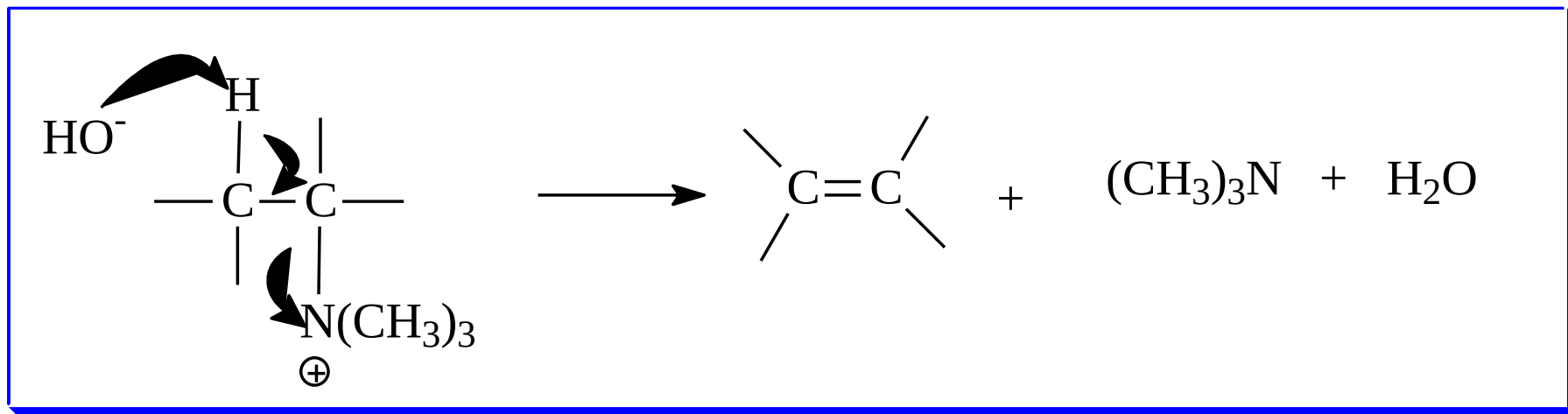
氨基在强酸作用下，由活化基团变成了钝化基团，因此最好将氨基保护后，再进行硝化、磺化反应



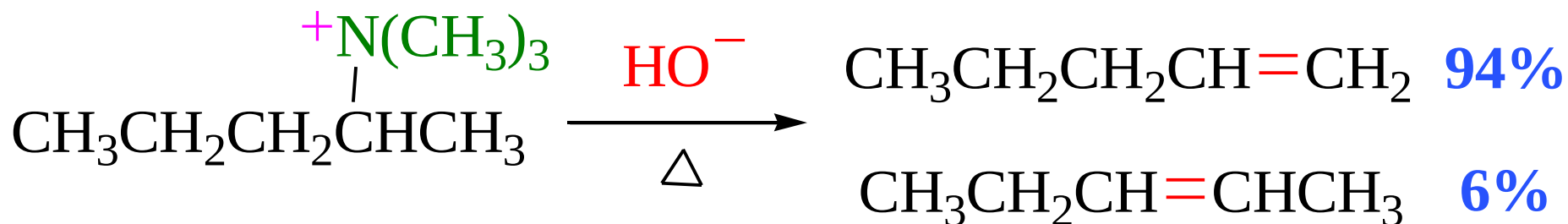
季铵盐化合物



季铵碱受热分解反应



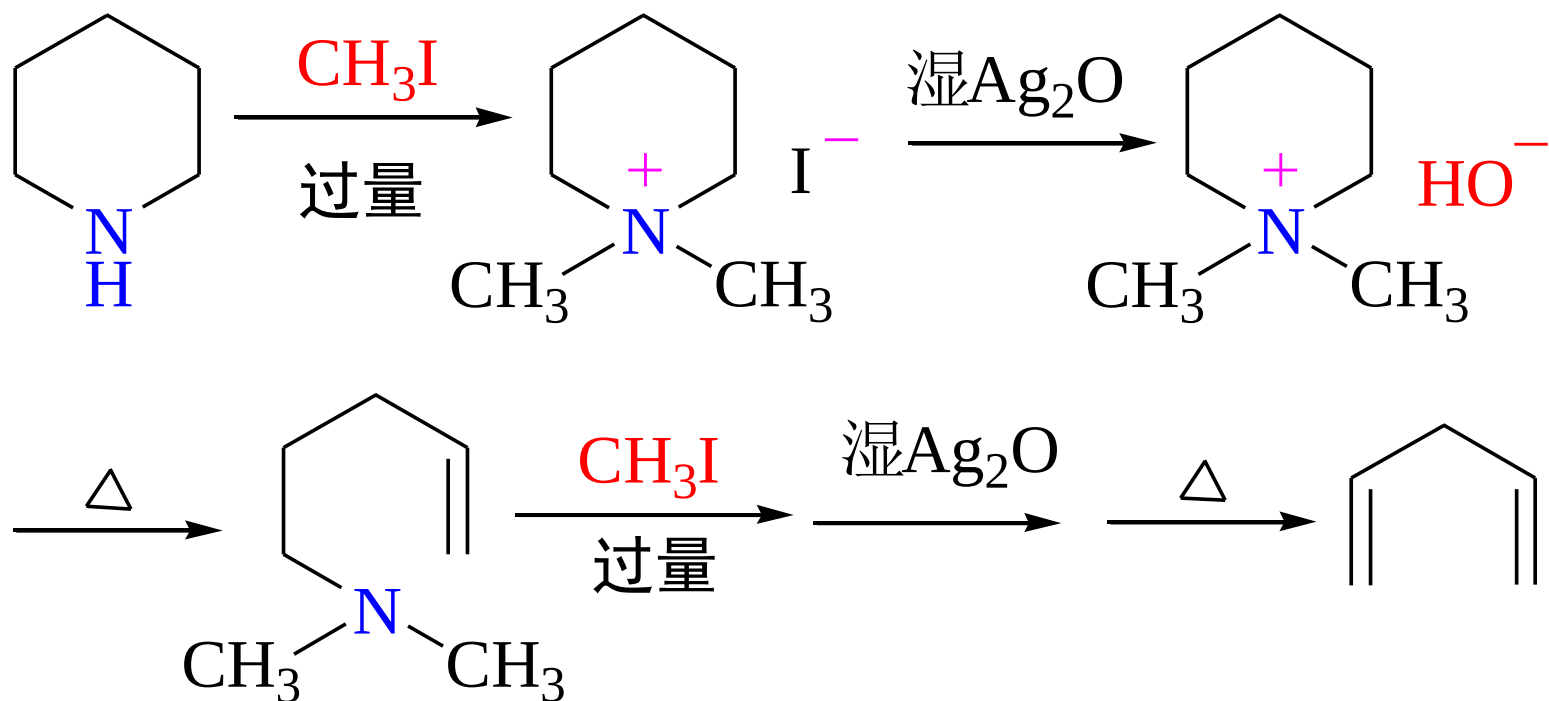
霍夫曼 (Hofmann) 消除



霍夫曼 (Hofmann) 消除

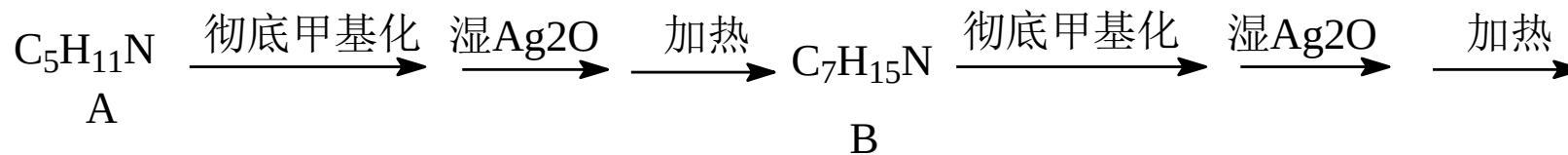
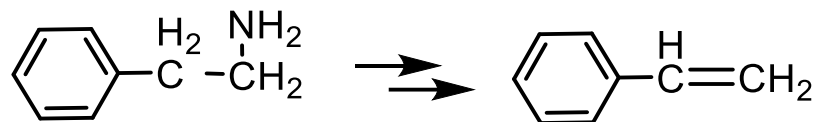
当季铵碱分子中有两个以上不同的氢原子可被消除时，反应主要从含氢较多的碳原子上消去氢原子，即主要生成双键碳原子上烷基取代较少的烯烃

应用：推测胺的结构



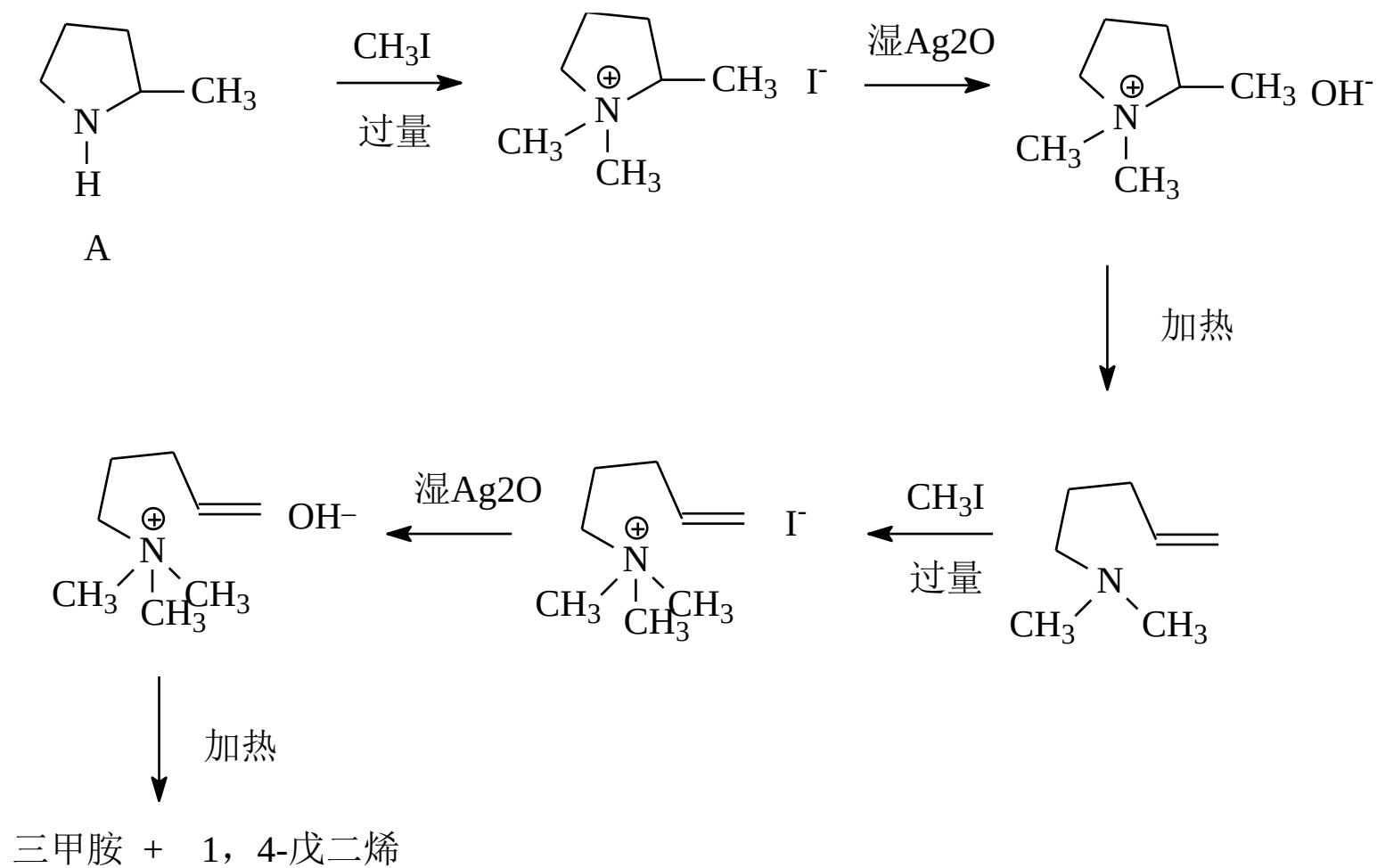
测试

完成下列转化



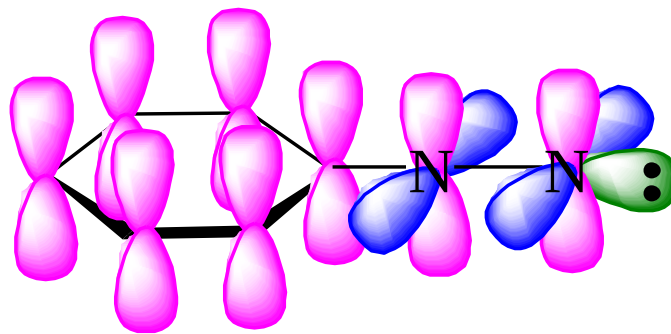
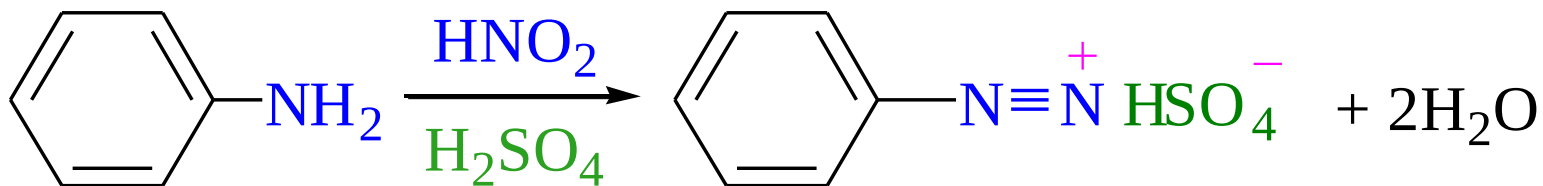
三甲胺 + 1, 4-戊二烯

请写出A和B的结构式



重氮与偶氮化合物

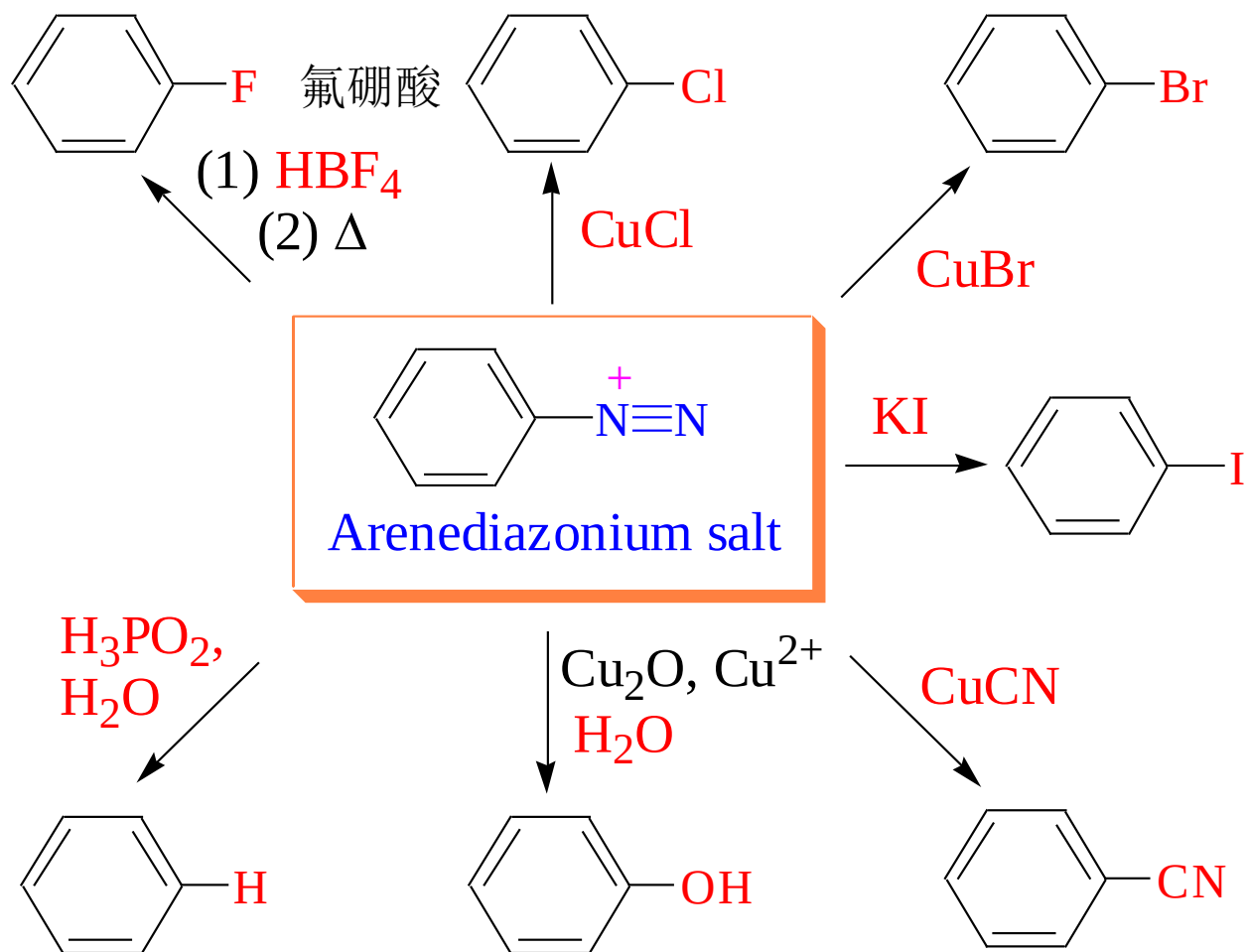
重氮盐的制备——重氮化反应



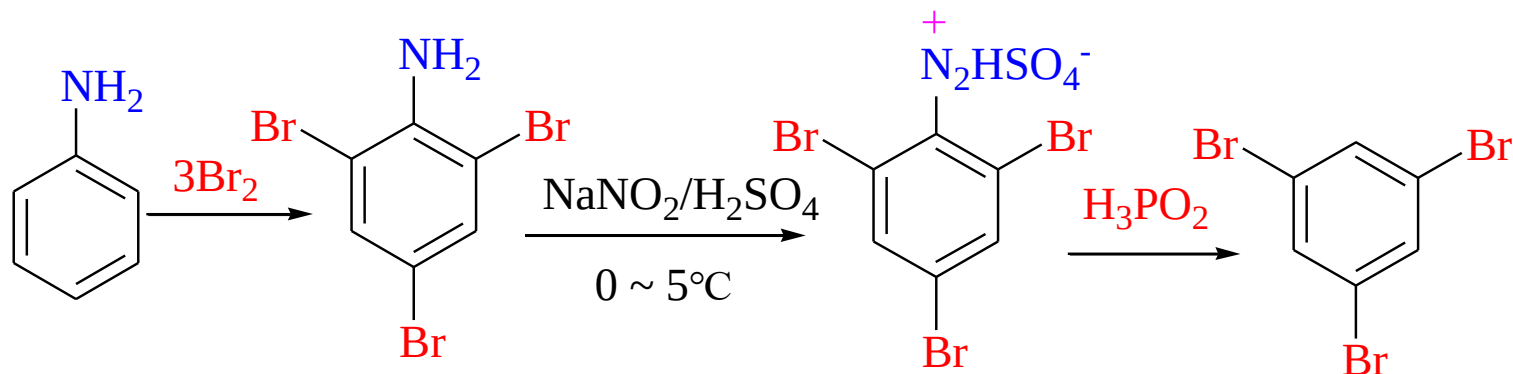
- 重氮盐是无色晶体，在干燥条件下不稳定，爆炸性强
- 重氮盐一般不溶于有机溶剂，可溶于水，其水溶液呈中性，在水溶液中发生离子化，溶液具有导电性
- 重氮化反应要在酸性条件下进行，酸的用量一般为1.5倍

芳香重氮盐在合成上的应用

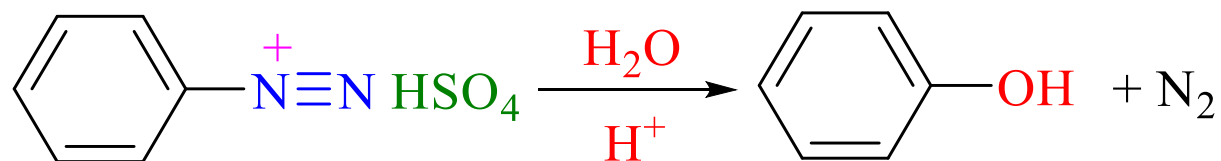
(1) 失去氮的反应



被氢所取代（去氨基反应）



被羟基所取代



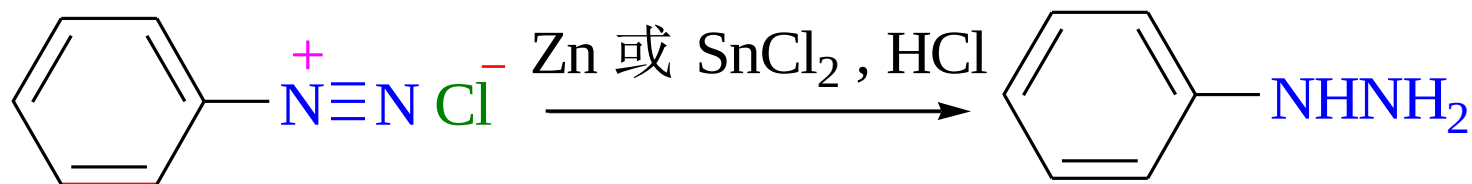
被卤素取代



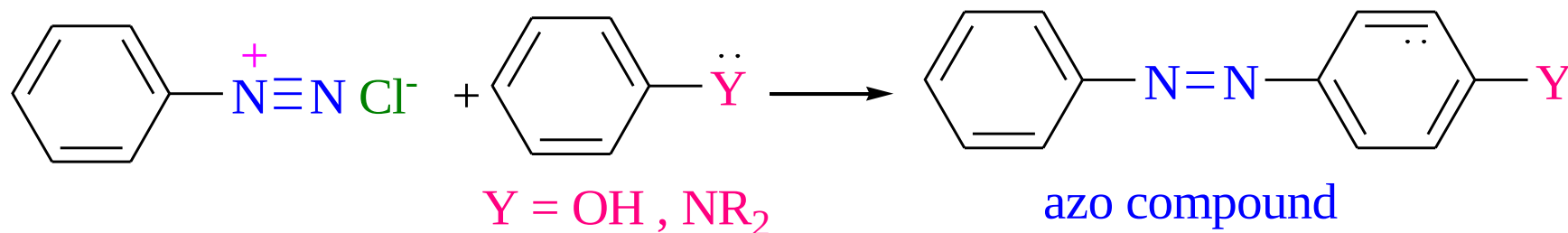
偶氮化合物制备

(2) 保留氮的反应

- 还原反应

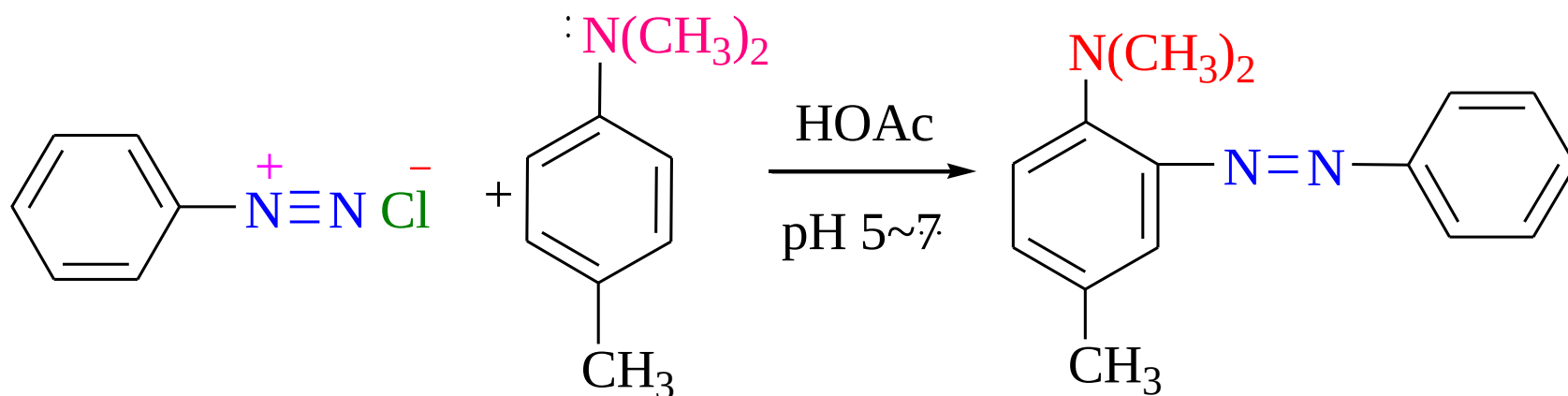


- 偶联反应

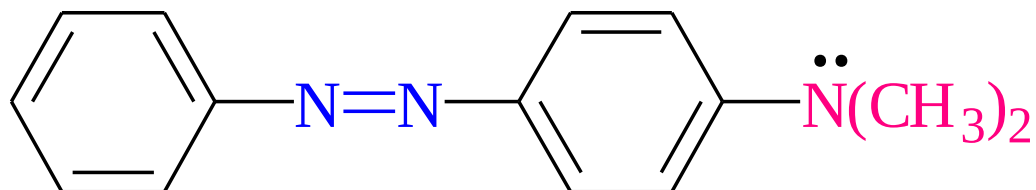
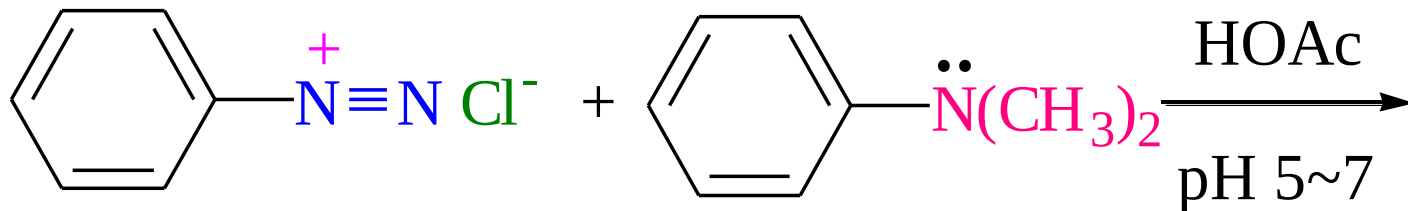
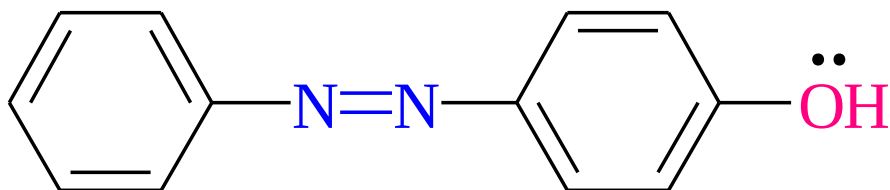
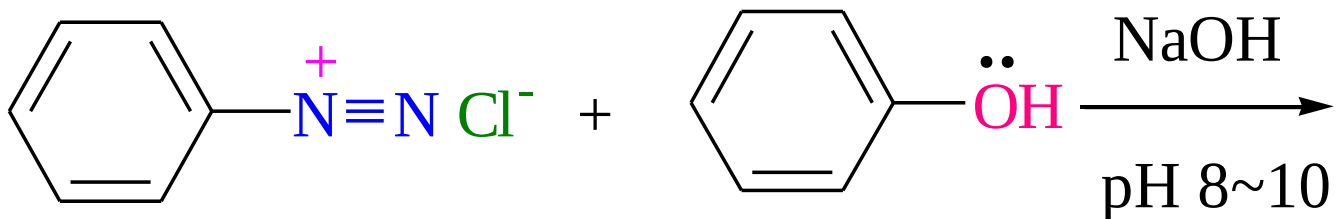


偶联反应

在微酸性中性或微碱性溶液中，重氮盐正离子作为亲电试剂可与连有强供电基的芳香族化合物，如酚芳胺等发生亲电取代反应生成偶氮化合物



Example



测试

请同学合成下列物质：

